Overlapping Prime-Shell Sign Transport Localizes a Finite Native-Budget Descent

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Abstract

TPC-303 found a finite obstruction to cardinality-only monotonicity of a source-first native weighted budget, but did not identify whether the associated signed target changes at the same location. We restrict the weighted labels generated by the TPC-302 physical-Gram enumeration to the overlaps of adjacent moving prime shells. After removing the independent global-sign gauge, the six fixed-source transport rows have mean correlations 1/2, 1/11, and 1/2 along the $Q = 50 \to 60$, $60 \to 70$, and $70 \to 90$ transitions. The middle transition is the unique low-correlation fracture. The TPC-303 budget census has descent counts $3, 15, 3$ and same-prefix descent counts $0, 9, 0$, so all nine same-prefix descents occur at that fracture. This is a finite localization certificate, not a causal separation or an asymptotic theorem.

1 Question and frozen data

TPC-302 established a source-first finite growing-grid gap between a signed weighted target and an all-positive control [2]. TPC-303 then tested the stronger shortcut that a larger shell cardinality should force a larger native weighted budget, and found 21 strict descents among 54 adjacent comparisons [1]. Its natural follow-up is to inspect the signed labels themselves rather than infer a mechanism from budget values alone.

We freeze $(N, H, z) = (512, 58, 5)$ and use the moving shells $Q = 50, 60, 70, 90$. Their cardinalities are $10, 13, 15, 17$. For each kernel exponent $e \in \{1, 2\}$, TPC-302 supplies a label $a_Q^{(e)} : S_Q \to \{-1, +1\}$, with the first shell label fixed to $+1$ only as a gauge convention. The shells are moving intervals, not a nested inclusion chain.

For adjacent shells put $O_{Q,Q'} = S_Q \cap S_{Q'}$. We compare labels only on this common prime set. This leaves the physical shell change and the source-first target change entangled, but makes that entanglement explicit.

2 Gauge-invariant transport identity

For binary labels a, b on a nonempty overlap O, define

$$\rho(a, b) = \max_{\varepsilon \in \{-1,+1\}} \frac{1}{|O|} \sum_{p \in O} a(p)\varepsilon b(p), \qquad d(a, b) = \min_{\varepsilon \in \{-1,+1\}} \frac{\#\{p : a(p) \neq \varepsilon b(p)\}}{|O|}.$$

Proposition 1 (Binary transport identity). *Writing $u = \sum_{p \in O} a(p)b(p)$, one has*

$$\rho(a, b) = \frac{|u|}{|O|}, \qquad d(a, b) = \frac{1 - \rho(a, b)}{2}.$$

Both quantities are invariant under independent global sign changes of a and b.

Table 1: Overlap transport and inherited budget census.

transition	overlap	ρ for $e=1, e=2$	mean ρ	fractures	budget desc.	same-prefix desc.
$50 \to 60$	$8, 8$	$1/2, 1/2$	$1/2$	0	$3/18$	$0/18$
$60 \to 70$	$11, 11$	$1/11, 1/11$	$1/11$	2	$15/18$	$9/18$
$70 \to 90$	$10, 10$	$3/5, 2/5$	$1/2$	0	$3/18$	$0/18$

Proof. The two possible signed inner products are u and $-u$, so the maximum is $|u|$. If m coordinates agree before alignment, the two possible mismatch counts are $|O|-m$ and m; the optimal one is their minimum. Since binary labels give $u = |O| - 2\min(m, |O|-m)$, division by $|O|$ yields the identity. Multiplying either label by a global sign changes u by at most a sign and does not change its absolute value or the minimized mismatch count. $\square$

The producer records the optimizing sign, all restricted labels, mismatch primes, and exact rational values. We call a row a coarse "fracture" when $\rho \le 1/3$. This threshold is declared for a readable finite crosswalk; the raw rational correlations are the primary data.

3 Finite crosswalk

Table 1 groups the six exponent/pair rows by adjacent Q transition. "Budget desc." is the number of TPC-303 descents among its 18 combinations of exponent, tolerance, and normalizer. The same-prefix column is the corresponding restricted count.

The middle overlap has aligned disagreement fraction $5/11$ for both exponents, while the other group means are $1/2$. Thus $60 \to 70$ is the unique minimum of the three group means and the only group with rows below the declared $1/3$ threshold. Independently, the TPC-303 transition census is $(3, 15, 3)$ for descents and $(0, 9, 0)$ for same-prefix descents. The unique minimum-correlation transition is therefore also the unique maximum-descent transition, and supports every same-prefix descent.

This statement is deliberately a crosswalk rather than a mechanism claim. The source-first label on each shell was generated from that shell's physical Gram matrix. A low overlap correlation can reflect a target-label switch, a physical operator change, or both. No counterfactual budget is evaluated here.

4 Obstruction and next question

The positive structural result is a gauge-corrected, exact finite transport observable whose minimum is localized without relying on the arbitrary first prime gauge. It also provides a compact bridge between two independently audited observables: label transport and native-budget descent.

The obstruction is equally important. Coincidence of minima does not imply that label switching causes the budget descent. To obtain that separation one must hold the physical shell/operator fixed while transporting a neighboring label, recompute the constrained native budget, and compare the counterfactual with the native target. Uniform profile-budget growth, arithmetic L^2, fixed-power credit, full Gate B, and a twin-prime conclusion remain open.

Reproducibility and scope

The canonical TPC-304 certificate locks the TPC-302 and TPC-303 producer and result hashes, stores all six overlap rows, and embeds the parent budget census. An independent checker reconstructs the rows without importing the producer; a stress suite exhausts small binary-label cases and global-sign flips. The Session-named Route-A/Route-B evaluator files are absent from this checkout, so no official evaluator pass is asserted.

References

[1] Liang Wang. A fixed-source cardinality-monotonicity obstruction for native prime-shell budgets, 2026. TPC-303 project release in the accompanying repository.

[2] Liang Wang. Source-first growing-shell stability of the native weighted/positive budget gap, 2026. TPC-302 project release in the accompanying repository.